MIL-S-52408B(ME)
19 July 1983
SUPERSEDING
MIL-S-52408A(ME)
20 October 1975

MILITARY SPECIFICATION

SPREADER, SLING: TELESCOPIC, VEHICLE LIFTING, 10-TON CAPACITY

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tubular-steel telescopic sling spreader.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

FF-P-386	- Pins, Cotter (Split).
FF-W-92	- Washer, Flat (Plain).
QQ-B-637	- Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
RR-C-271	- Chains and Attachments, Welded and Weldless.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS20392	- Pin, Straight, Headed - Drilled Shank.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM A 36 - Structural Steel.
ANSI/ASTM A 322 - Steel Bars, Alloy, Standard Grades
ANSI/ASTM A 513 - Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing.
ANSI/AWS A2.4 - Symbols for Welding and Nondestructive Testing.
ANSI B46.1 - Surface Texture (Surface Roughness, Waviness And Lay).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The sling spreader shall be as shown in figures 1 through 4, and as specified herein. When assembled, the sling spreader shall be adjustable from 7 to 10 feet in 3 inch increments for locking in position with toggle pins.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The sling spreader shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Tubing. Steel tubing shall conform to ANSI/ASTM A 513, grade 4130, type and condition optional.

3.3.3 End plates. End plates shall be steel conforming to ANSI/ASTM A 322, UNS G41300, condition optional.

3.3.4 Toggles. Toggles shall be brass conforming to QQ-B-637, copper alloy No. 464, half-hard temper.

3.3.4.1 Toggle pins. Toggle pins shall be steel conforming to ANSI/ASTM A 36.

3.3.5 Chain. Steel chain shall conform to RR-C-271, type II, class 4, nominal thickness 0.065 inch.

3.3.6 Flathead pins. Steel flathead pins shall conform to MS20392, dash No. 12C109.

3.3.7 Washers. Steel washers shall conform to FF-W-92, type A, grade I, class A, natural finish.

3.3.8 Cotter pins. Cotter pins shall conform to FF-P-386, type B, class optional.

3.3.9 Finish. Surface and edge finish as shown in figure 2, 3, and 4 shall be in accordance with ANSI B46.1.

3.4 Performance. The assembled sling spreader shall lift a safe working load of 20,000 pounds and a proof load of 40,000 pounds without rupture or permanent deformation of any component.

3.5 Identification marking. The spreader shall be identified in accordance with MIL-STD-130. Markings shall be 0.25 inch high and shall be die-stamped on the outer side of the center section.

3.6 Treatment and painting. The portions of the spreader normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color of the finish paint shall be forest green.

3.7 Workmanship. Components of the spreader shall not have burrs or projections which could prove injurious to using personnel. The spreader components shall not have cracks, splits, or breaks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

3.7.1 Welding. The surface of all parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall be free from defects such as porosity, splatter, entrapped slag, overlap, undercut, nonfusion, and cracks. Welding shall not be used as a repair measure. Interpretation of welding symbols in figures 1 and 3 shall be in accordance with ANSI/AWS A2.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article sling spreader shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article sling spreader shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level S-2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for the minor defect.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective

4.5 Inspection schedule.

4.5.1 Examination. The sling spreaders shall be examined as specified herein for the following defects:

Major

101. Dimensions or tolerances not as specified.
102. Materials not as specified.
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Components missing or not as specified.
107. Treatment and painting not as specified.
108. Workmanship not as specified.

Minor

201. Identification marking not as specified.

4.5.2 Proof load test. Extend the sling spreader to its maximum length and attach a 2 leg wire rope sling through the holes in the end plates. The two upper sling legs shall be of equal length and shall converge in a lifting ring or shackle. During application of the proof load, the angles between each of the upper sling legs and the spreader shall be 45 degrees, plus or minus 2 degrees, and the angle between each of the lower sling legs and the spreader shall be 90 degrees, plus or minus 2 degrees. The two lower sling legs shall be of equal length and shall be subjected to a gross weight of not less than 40,000 pounds. Apply a vertical hoisting force to the lifting ring or shackle and raise the spreader with the proof load until the load is free of any support. Raise and lower the loaded spreader to the floor five times and hold suspended for periods of 5 to 10 minutes for a total of 30 minutes. A hydraulically applied proof load may be substituted for the gross weight of equal force. After the proof load test, examine the spreader for evidence of rupture or deformation. Nonconformance to 3.4 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

109. Spreaders not reduced to minimum size as specified for level A or B.
110. Boxes not as specified for level A or B.
111. Quantities packed together exceed the weight limitation of the box for level A or B.
112. Packing not in accordance with the referenced document for commercial.
113. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.1.1 Level A. The spreaders shall be reduced to minimum size without disassembly. The spreaders shall then be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitation of the box. The box shall be closed and strapped in accordance with the appendix to the applicable box specification.

5.1.2 Level B. The spreaders shall be packed as specified in 5.1.1 for level A, except boxes shall be domestic type or class, as applicable.

5.1.3 Commercial. Commercial packing of the spreaders shall be in accordance with MIL-STD-1188.

5.2 Marking (see 6.5).

5.2.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packing shall be in accordance with MIL-STD-1188.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The spreader is intended for use with wire rope slings employed to lift vehicles weighing up to 10 tons and is intended to eliminate contact between the sling and the sides of the vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Degree of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first produced model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Packaging. Degrees of preservation have not been included as the item needs no preservation. For purposes of preservation/packing level marking the preservation shall be considered level A.

Custodian:
Army - ME

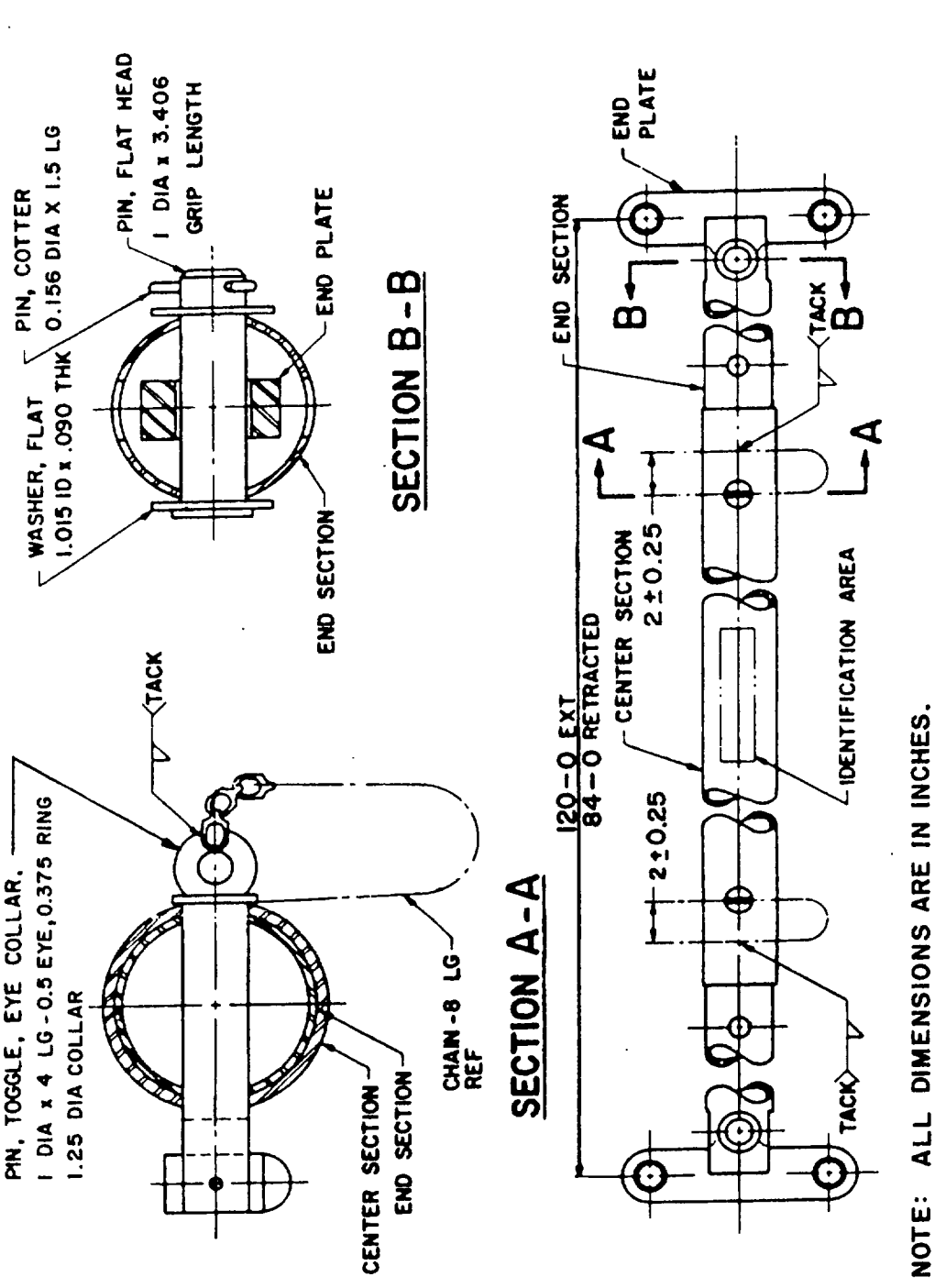
Review activities:
Army - SM
DLA - IS

User activity:
Army - MT

Preparing activity:
Army - ME

Project 3940-A030

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

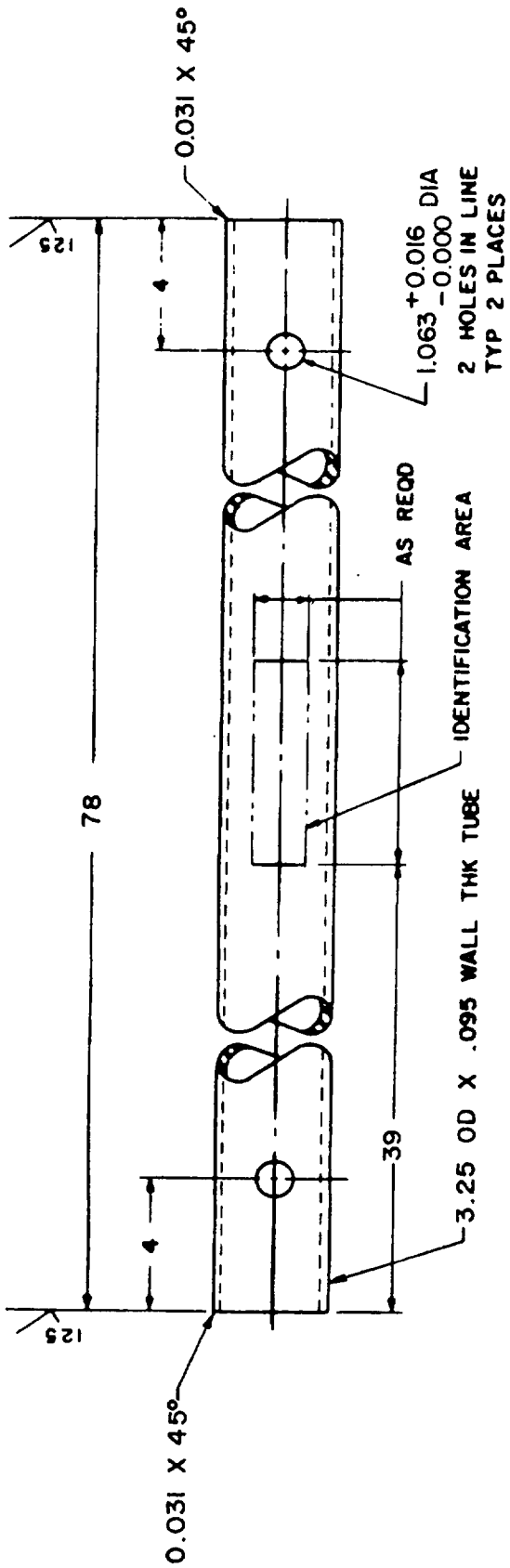


NOTE: ALL DIMENSIONS ARE IN INCHES.

FIGURE 1. Telescopic spreader bar assembly.

X-2901A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.



NOTES:
1. TOLERANCE UNLESS OTHERWISE SPECIFIED ± 0.062.
2. ALL DIMENSIONS ARE IN INCHES.

X-2902A

FIGURE 2. Center section.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

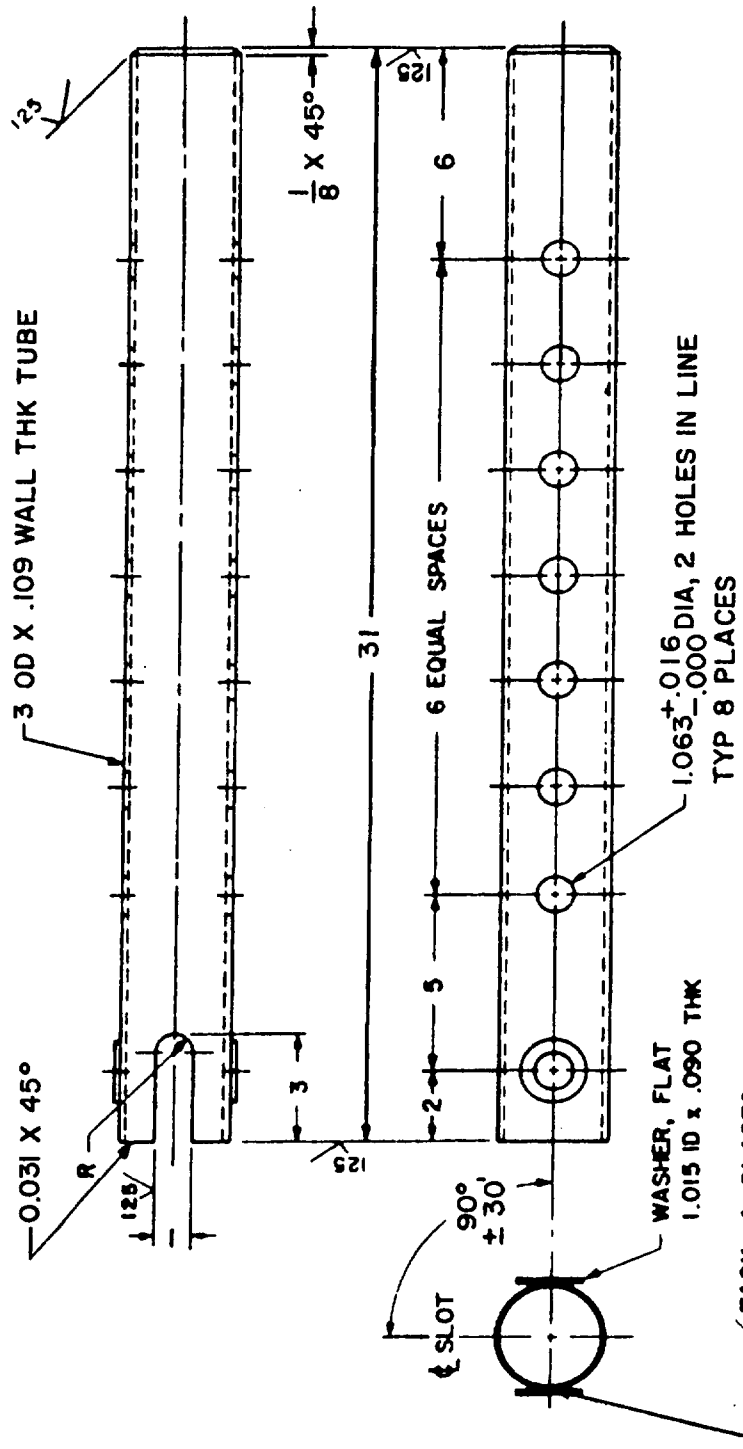
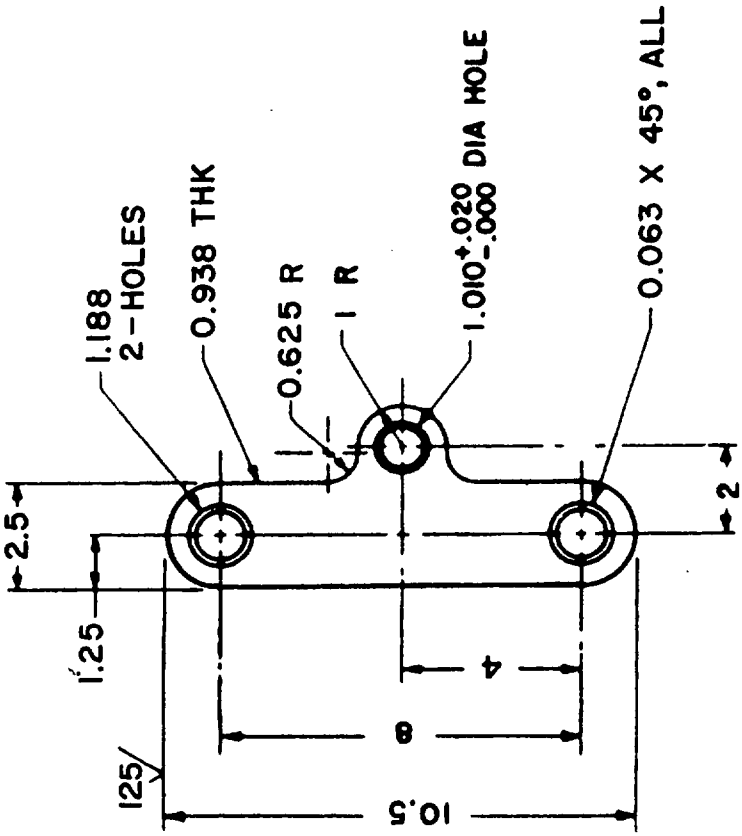


FIGURE 3. End section.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.



NOTES:
1. TOLERANCES UNLESS OTHERWISE SPECIFIED ± 0.063, ANGLES ± 1°.
2. ALL DIMENSIONS ARE IN INCHES.

X-2904A

FIGURE 4. End plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use submitting comments and suggestions for improvements. All users of military standardization documents are invited to provi suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, an mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, w too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

USAMERADCOM
DRDME-DS
Ft. Belvoir, VA 22060

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment Research and
Development Command
DRDME-DS
Fort Belvoir, VA 22060

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-52408B(ME)	Spreader, Sling: Telescopic, Vehicle Lifting, 10-Ton Capacity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:17Z
Check the source to verify that this is the current version before use.